

pandoc-linguexx

Converting linguexx documents to LibreOffice Writer and Word

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Abstract

`pandoc-linguexx` converts a $\text{\LaTeX}$ document written with `linguexx` into a LibreOffice Writer (`.odt`) or Word (`.docx`) file in which every example number is a live field. Insert an example in the word processor and everything after it renumbers, cross-references included. The rest of the document — sections, prose, footnotes, tables, citations — goes through `pandoc`, and the converter finishes what `pandoc` leaves undone: numbered headings, captions and equations, and cross-references to all of them, live fields wherever the word processor allows. The project also contains a Writer macro and Word and OnlyOffice add-ins that build the same examples inside the word processor, and a converter for the way back to $\text{\LaTeX}$.

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1 Introduction

A linguist who writes in `LATEX` with `linguexx` sooner or later has to hand a paper to someone who works in a word processor: a co-author, an editor, a journal that accepts only `.docx`. Pandoc converts the prose well, but it knows nothing of glossed examples: the alignment of words and glosses is lost, and the example numbers become plain text that no longer follows when an example is added.

`pandoc-linguexx` keeps both. Each example becomes a borderless table whose columns align the object language with its glosses, and each number is a field — a *number range* field in Writer, a `SEQ` field in Word — so that the word processor rennumbers. References to examples are fields pointing at those numbers.

The project has three parts, which share one specification:

`linguexx2odt`

the converter: `LATEX` to `.odt` or `.docx`. Most of this manual is about it.

The Writer macro and the Word and OnlyOffice add-ins

build the same examples inside the word processor from a few typed lines, so that a converted document can be extended in kind (section 8).

`docx2linguexx` and `odt2linguexx`

take a document edited in the word processor back to `LATEX` (section 9).

Four principles run through all of it:

1. **Numbers are live.** An example number, a section number, a caption's number and every reference to them is a field the word processor recomputes. Where that is not possible, the manual says so.

2. **Nothing is direct formatting.** Every paragraph and character the converter writes carries a named style (`LxExampleCell`, `LxLeipzig`, ...), so a whole document is restyled from the style list.
3. **L^AT_EX is the reference.** What a reference, a caption or a number prints is not reasoned out: it was measured from what L^AT_EX prints for the same source, and the test suite holds the converter to it.
4. **It never fails silently.** Anything it cannot render faithfully is named in a warning. Some constructs degrade, a few are deleted; section 10 lists them.

Note. The `.odt` target is the reference one: it is what the Writer macro is built against. The `.docx` target writes the same examples, fields and styles, and reads correctly in Word 365, LibreOffice and OnlyOffice.

2 Installation

2.1 Requirements

Program	Version	Needed for
pandoc	3.0 or later	every conversion; tested with 3.6.1 and 3.10.2
Python	3.10 or later	the converter itself
fonttools	any	measuring a face other than Times (installed with the package)
LibreOffice	optional	the page numbers of <code>\pageref</code> (section 6.8), <code>odt2linguexx</code> , the test suite
pdfinfo (poppler)	optional	the page numbers, with LibreOffice
kpsewhich (T _E X)	optional	finding files <code>\input</code> names through TEXINPUTS

LibreOffice is *not* needed to convert. It is only needed to read the result, to fill in page numbers ahead of time, and to run the half of the test suite that renders documents.

2.2 Installing

The package installs the three commands `linguexx2odt`, `docx2linguexx` and `odt2linguexx`. With `pipx`, the command lands on the `PATH` by itself:

```
| pipx install -e /path/to/pandoc-linguexx
```

Many current distributions refuse `pip install` into the system Python. Use a virtual environment instead:

```
| python -m venv .venv
|.venv/bin/pip install -e .
|.venv/bin/linguexx2odt paper.tex
```

Or install nothing, and run it from a checkout:

```
| PYTHONPATH=src python3 -m linguexx2odt paper.tex
```

The `PYTHONPATH` is not optional there: the project uses a `src` layout, so a bare `python3 -m linguexx2odt` in the repository's root does not find the package.

3 Quick start

```
linguexx2odt paper.tex          # writes paper.odt
linguexx2odt paper.tex --to docx # writes paper.docx
linguexx2odt paper.tex -o draft.odt -v # another name; report what was done
```

The run prints a warning for anything it could not render faithfully, with the line it is about. Read them: they are the list of what to check in the output.

Figure 1 shows three examples as `linguexx` sets them and as the converter writes them. The source is the same, given below the figure.

L^AT_EX with `linguexx`

- (1) The cat is on the mat.
- (2)
 - a. Le chat dort
the cat sleeps
'The cat sleeps.'
 - b. *Chat le dort
cat the sleeps
'The cat sleeps.'
- (3) Hans hat DEM Kind ein Buch gegeben
Hans has the.DAT child a book given
'Hans gave the child a book.'

Converted to `.odt`, as LibreOffice lays it out

- (1) The cat is on the mat.
- (2)
 - a. Le chat dort
the cat sleeps
'The cat sleeps.'
 - b. *Chat le dort
cat the sleeps
'The cat sleeps.'
- (3) Hans hat DEM Kind ein Buch gegeben
Hans has the.DAT child a book given
'Hans gave the child a book.'

Figure 1: The same source set by L^AT_EX (above) and converted to `.odt` (below). Numbers, sub-example letters, the judgment hanging to the left, glosses under their words and `{ein Buch}` as one column all survive.

```
\ex.\label{ex:basic} The cat is on the mat.

\ex.\label{ex:para} \a. \gll Le chat dort\\
                    the cat sleeps\\
                    \glt `The cat sleeps.'
\b. * \gll Chat le dort\\
                    cat the sleeps\\
```

```

\glt `The cat sleeps.'

\ex. \gll Hans hat \lpzg{dem} Kind {ein Buch} gegeben\\
      Hans has the.\lpzg{dat} child a book given\\
\glt `Hans gave the child a book.'
```

To check that the numbers are live: in Writer, put the cursor before an example, insert a new number field (Insert → Field → More Fields → Variables → Number range, NumEx), then update the fields (Tools → Update → Fields, or F9). Every later number and every reference shifts. The Writer macro (section 8) does this for you, building the whole example around the number.

4 The command line

```
| linguexx2odt [options] input.tex
```

Option	Effect
<i>Output</i>	
<code>-o, --output <i><file></i></code>	The output file. Default: the input's name with the target's extension.
<code>--to odt docx</code>	The target format. Default: odt.
<code>--reference-doc <i><file></i></code>	A pandoc reference document supplying the base styles. It also fixes the face (<code>--font</code> is then ignored). Its heading numbering is replaced by the converter's (section 6.3).
<code>--page a4 a4-wide letter keep</code>	The page geometry written into the .odt. Default: a4; keep leaves pandoc's US Letter.
<i>Layout of the examples</i>	
<code>--text-width <i><cm></i></code>	The width of the text block the columns are computed for. Default: 17 (A4 with 2 cm margins).
<code>--font <i><name></i></code>	The face the document is set in and its columns are measured for. Default: Times New Roman (section 5.5).
<code>--font-pt <i><pt></i></code>	The body size assumed for the measurements. Default: 12.
<code>--no-split</code>	Do not break a glossed example too wide for the block into bands; squeeze it into one instead (section 5.3).
<code>--example-spacing <i><cm></i></code>	The space above and below each example. Default: 0.18. Afterwards it is the style LxExampleSpace.
<code>--space-above <i><cm></i>,</code>	Override it above or below only (styles LxExampleSpaceAbove, LxExampleSpaceBelow).
<code>--space-below <i><cm></i></code>	
<i>Citations</i>	
<code>--bibliography <i><file.bib></i></code>	A bibliography to resolve citations against; repeatable. Default: those the document names with \addbibresource or \bibliography.
<code>--csl <i><style></i></code>	A CSL style. Default: pandoc's Chicago author–date, which reads like natbib's author–year mode.
<code>--no-citeproc</code>	Do not resolve citations; print their keys.
<i>Diagnostics</i>	

Option	Effect
<code>-q, --quiet</code>	Suppress the warnings.
<code>-v, --verbose</code>	Report what was converted.
<code>--keep-intermediates</code> <i><dir></i>	Keep the intermediate files: the residue pandoc reads, its AST, the file before post-processing.
<code>--print-macro</code>	Print the Writer macro's source and exit.

5 Examples

5.1 What is converted

The converter reads `linguexx`'s *lazy* syntax, which is `linguex`'s:

- `\ex.`, sub-examples `\a.`, `\b.`, ... on two levels (letters, then roman numerals), `\z.` to end a level early;
- glosses: `\gll`, `\glll`, `\gl ... \endgl`, the translation `\glt`, the shorthands `\exg.`, `\ag.`, `\bg.`;
- `{braced groups}` as one column, tiers of unequal length;
- judgments `(*, ??, \#, \jdg{...})`;
- `\label`, `\sublabel`, `\ref`, `\pref`, `\cref`, the relative references `\Next`, `\Last`, `\NNext`, `\LLast`, ranges `\refrange` and `\prefrange` (section 6);
- `\lpzg{...}` for Leipzig gloss labels, in small capitals;
- the brackets `\ExLBr/\ExRBr`, the counters `\Exalph` and `\Exroman`, structural labels `\exannot{...}` in a column at 0.75 of the text block, as `\ExAnnotColumn` puts them;
- custom labels `\ex. [(4')]`, printed as they are, and inline verbatim (`\verb`, `\Verb`, `\lstinline`);
- footnotes inside an example, and the document's own macros, which are expanded.

An example ends where `linguexx` ends it: at a blank line, at `\z.`, at an environment boundary or at a closing brace.

5.2 How an example is laid out

Each example is one borderless table, so its columns line up:

- The number is a field (`NumEx` in both formats) between literal brackets: the brackets are the document's, `\ExLBr` and `\ExRBr`.
- Each object-language word sits in a cell of its own, with its glosses in the cells below it. The free translation spans the last row.
- A judgment mark *hangs* to the left of the text, in a column cut out of the one before it. A judged and an unjudged example therefore begin at the same place, as in `linguexx`, and a sub-example's letter sits where a main example's text begins.
- A paradigm of sub-examples shares one table, but each item is measured on its own words: the table's columns are the union of every boundary its items need.

5.3 Long examples: bands

A glossed example too wide for the text block is split into *bands*: groups of tier rows, each starting again at the left margin (figure 2). The number appears on the first band only

and the translation once, at the end. With `--no-split` the example is squeezed into a single band instead, its words wrapping inside their cells.

- (1) Ce matin-là, les enfants du village voisin sont descendus jusqu'à la
this morning-there the children of the village neighbouring are gone down until to the
rivière pour regarder passer les bateaux
river for watch pass the boats
‘That morning, the children from the neighbouring village went down to the river to watch
the boats go by.’

Figure 2: A glossed example too wide for the text block, split into two bands. Each band starts again at the left; the translation comes once.

A word processor's table does not reflow. The break points are computed at conversion time from `--text-width` and then fixed. If the page changes later, convert again from the `.tex` with the new width, or let the Writer macro rebuild the example against the page it is on.

5.4 Named styles

Everything the converter writes carries a named style. Change the style, and every example in the document follows.

Style	What it is
<i>Paragraph styles</i>	
LxExampleCell	A cell of an example: no space before or after, no indent.
LxTranslation	The free translation, with a little space above.
LxJudgmentCell	The judgment mark, right-aligned against the text it judges.
LxAnnot	The column of <code>\exannot</code> labels.
LxExampleBand	Marks the first row of a continuation band; it looks like LxExampleCell.
LxExampleSpace	The space above and below each example: an empty spacer row at a fixed line height. Change this one to move every gap at once.
LxExampleSpaceAbove	Inherits it; give it a height of its own to change the gap above only.
LxExampleSpaceBelow	The same, below.
LxExampleGap	<code>.docx</code> only: a 1 pt paragraph between two consecutive examples, because Word joins tables that touch.
LxEquation	A numbered display equation: a centre and a right tab stop (section 6.6).
<i>Character styles</i>	
LxLeipzig	A Leipzig gloss label, <code>\lpzg</code> : small capitals.
LxSmallCaps, LxItalic, LxBold, LxUnderline	<code>\textsc</code> , <code>\textit</code> and <code>\emph</code> , <code>\textbf</code> , <code>\underline</code> inside an example.
LxSuperscript, LxSubscript	Raised and lowered text inside an example.
LxJudgment	A judgment mark.

To change a gap, edit the style's Indents & Spacing → Line spacing → Fixed in Writer. A fixed line spacing sets the height exactly; 0 cm really is zero.

5.5 Column widths and the face

A word processor sizes a table's columns from the widths the file gives, and the converter has to compute them without a word processor to ask. It does so from per-character measurements of the face. The face the document is set in and the face its columns are measured for must therefore be the same one, and `--font` sets both:

```
| linguex2odt paper.tex --font "EB Garamond"
```

Times New Roman is the default. Faces metric-compatible with it (Liberation Serif, Nimbus Roman, Tinos) use a table measured once and shipped with the converter. Any other face is measured from its font file with `fonttools`, in milliseconds. A name that no installed font answers to is an error, not a silent substitution.

The estimate errs slightly wide, by -2% to $+9\%$ of the rendered width. Text is measured as it is *drawn*: a gloss label in `\lpzg` is set in small capitals, which are wider than the lowercase they replace, and is measured as such. Set `--font-pt` if the body text is not 12pt. Supplying a `--reference-doc` is choosing a face. In that case `--font` is ignored and the columns may be a little off.

5.6 The .docx target

- **Every field is written with its correct value already in it.** Word and LibreOffice recompute fields when a document is opened or printed; OnlyOffice shows what is stored. Both read right.
- The file is validated against the ECMA-376 transitional schemas (`make schemas`, then `tools/validate_docx.py`), and Word 365 opens it without offering to repair it.
- Consecutive examples are kept apart by a 1 pt paragraph in the style `LxExampleGap`: Word joins tables that touch.

6 Cross-references

A $\text{\LaTeX}$ document refers to examples, sections, tables, figures, footnotes, list items, equations and pages. The converter turns each reference — except one to an equation, which is text (section 6.6) — into a field of the word processor that points at the thing referred to. That thing is numbered by the word processor too, so moving a section or inserting a table renumbers it and every reference to it. Figure 3 shows a page of them.

What a reference prints was measured against $\text{\LaTeX}$ with `hyperref` and `cleveref`, for every command and every kind of target. A reference prints what $\text{\LaTeX}$ prints, with the few exceptions in section 6.10.

6.1 What each reference becomes

1 Data

(1) An example.

Table 1: Two words.

form	gloss
chat	cat
$[[cat]] = \lambda x. \text{cat}(x)$	

(1)

1.1 Analysis

1. First point.

2. Second point.

See (1), section 1.1, table 1, (1), items 1 and 2; sections 1 and 1.1 start on page 1.

Figure 3: Headings, a caption, an equation and a list, numbered as L^AT_EX numbers them, and references to each, all of them fields but the equation's. The source is `figures/src/refs.tex`; L^AT_EX prints the same last line.

Target	.odt	.docx
example	<code>text:sequence-ref</code> to the NumEx	REF to a bookmark around the SEQ
section	<code>text:bookmark-ref</code> , format <i>chapter</i>	REF \w, the number in full con- text
section title (\nameref)	<code>text:bookmark-ref</code> , format <i>text</i>	REF, the bookmarked title
table, figure	<code>text:sequence-ref</code> to the caption's sequence	REF to a bookmark around the caption's SEQ
footnote	<code>text:note-ref</code>	NOTEREF
list item	one <code>text:bookmark-ref</code> per level, format <i>number- no-superior</i>	one REF \n per level
equation	text	text
page	the reference's <i>page</i> format	PAGEREF

Each field is written with the value L^AT_EX prints already in it. That is what a reader that never updates fields shows, and it is right, pages included (section 6.8).

6.2 Examples

`\ref{x}` prints “(3)”, `\pref{x}` the bare “3”, and a sub-example “(3a)”. The brackets are the document's `\ExLBr` and `\ExRBr`. References may point forward, and they may appear inside examples. The relative references `\Next`, `\Last`, `\NNext`, `\LLast` and their `p` forms are resolved by position, as `linguexx` resolves them; inside an example `\Last` is the example itself. Ranges, `\refrange` and `\prefrange`, print “(3a–c)” or “(1–3a)” with a live field at each end. `\cref` to an example prints “(1)”, as `linguexx` sets it up. A label that no example carries prints “??”, with a warning.

A reference to an example with a *custom* label, `\ex. [(7)]`, prints the label, “(7)” or “(5a)”, as `linguexx` 1.4 does. It is text, not a field: a custom label is not a counter, and nothing rennumbers it.

6.3 Sections and the appendix

The converter numbers the headings, because pandoc does not: `\section`, `\subsection` and `\subsubsection` as “1”, “1.1”, “1.1.1”, by the word processor’s outline numbering, so they renumber when a section is moved. A `\section*` gets no number and does not advance the count. A reference to it, or to a `\paragraph`, prints the last number set before it, as $\text{\LaTeX}$ does.

Command	Prints
<code>\ref{s}</code>	1.1
<code>\cref{s}</code> , <code>\Cref{s}</code>	section 1.1, Section 1.1
<code>\autoref{s}</code>	subsection 1.1 (the name follows the level)
<code>\nameref{s}</code>	the heading’s title
<code>\pageref{s}</code> , <code>\cpageref{s}</code>	3, page 3

After `\appendix` the sections are lettered, “A”, “A.1”, in the headings and in every reference to them. `\autoref` says “Appendix A” in the document’s language. Pandoc drops `\appendix` without a trace, so the converter marks its place before pandoc reads the document. The appendix’s headings take a numbering of their own: a list in Writer, a second numbering definition in Word.

The converter’s heading numbering replaces any that a `--reference-doc` defines.

6.4 Tables, figures and footnotes

Pandoc writes a caption without a number. The converter numbers it as $\text{\LaTeX}$ does, “Table 1: A table.”, with a live sequence for tables and another for figures. Only a float with a `\caption` is counted, as only `\caption` advances $\text{\LaTeX}$ ’s counter. `\ref` prints “1”, `\cref` “table 1” or “fig. 1”, `\autoref` “Table 1”, and `\nameref` the caption without its final full stop.

A reference to a footnote points at the note’s mark in the text and shows its number: “footnote 2”. A footnote inside an example is a real footnote, in both formats, and is counted with the rest. A `\label` inside a footnote belongs to the footnote, as in $\text{\LaTeX}$, even when the footnote is inside an example or a list item.

6.5 Lists

The converter numbers an `enumerate` as the `article` class does, “1.”, “(a)”, “i.”, “A.” by depth, from its start number (so `\setcounter{enumi}{4}` gives “5.”). A list whose style its source names — the `enumerate` package’s `[(i)]` — keeps that style.

A reference to an item is a field per level, on the item and each of its parents: “3”, “2(b)i”. Pandoc writes a nested list as a list of its own, so no word processor’s “number in context” knows the parents. `\cref` prints “item 3”, and `\nameref` the title of the section the item is in, as $\text{\LaTeX}$ does.

6.6 Equations

Pandoc writes display equations with no number. The converter numbers them as `amsmath` does, and sets each number flush right beside its equation. The paragraph style `LxEquation` has a centre and a right tab stop, which is how Writer and Word number their own equations.

- `equation` and `multline`: one number;

- `align`, `gather`, `alignat`, `flalign`, `eqnarray`: one number per row, none for a row with `\nonumber` or `\notag`;
- `\tag{x}`: “(x)”, not advancing the count;
- starred environments, `\[...]` and `displaymath`: none;
- `\numberwithin{equation}{section}` (or `\counterwithin`): “(1.2)”, “(A.1)”, starting again at each section.

An `align` with several numbers is set one row per line, so that each number stands beside its row; its & alignment is lost there. One with a single number stays whole.

A reference to an equation is **text**: “1”, `\eqref` “(1)”, `\cref` “eq. (1)”, `\Cref` “Equation (1)”, `\autoref` “Equation 1”. Neither the numbers nor the references follow a renumbering in the word processor. `\nameref` is the live title of the section the equation is in, and `\pageref` a live page.

6.7 Several labels: `\cref` lists and ranges

`\cref` and `\Cref` with several labels, and `\crefrange`, print what `cleveref` prints:

- the labels go in groups by kind, in the order each kind first comes: `\cref{s1,t1,s2}` is “sections 1 and 2 and table 1”;
- within a group they are sorted, a repeat is dropped, and a run of three or more consecutive numbers becomes a range: “sections 1 to 3 and 5”;
- groups are joined as `cleveref` joins them: “section 1, table 1, and eq. (1)”, with or without the last comma depending on the language;
- a group of several takes the plural name, and `\Cref` capitalises the first group only.

Each number is a field of its own. The grouping, sorting and ranges are worked out at conversion time; they stay as they are when the numbers change later.

6.8 Pages

A page is known only once the document is laid out, and a converter does not lay out. So when LibreOffice and `pdfinfo` are installed, the converted file is laid out once: it is exported to PDF with its bookmarks as named destinations, and each page field gets the page its bookmark lands on. Word and LibreOffice still compute their own when they update the fields. A reader that never updates them, such as OnlyOffice with a `.docx`, shows LibreOffice’s page.

Without LibreOffice a page field shows “?” until the word processor updates it, and the run says so once. In an `.odt`, a page reference to an *example* or a *footnote* always shows “?” until then, since neither is a bookmark the layout can report; LibreOffice computes it when it loads the file.

6.9 The language of the names

“section 1” is “Abschnitt 1” in a German document, and a caption’s “Table 1:” is “Tableau”—or rather “Table 1 —” in French. Three packages supply these names, and each has its own idea of the document’s language. The converter follows them as $\text{\LaTeX}$ does:

Names	Follow
<code>\cref</code> , <code>\Cref</code> , <code>\cpageref</code>	the language given to <code>cleveref</code> as an option, or to the class. <i>Not</i> <code>babel</code> 's option alone: with <code>\usepackage[ngerman]{babel}</code> and a bare <code>\usepackage{cleveref}</code> , $\text{\LaTeX}$ prints “section 1”, and so does the converter.
<code>\autoref</code>	<code>babel</code> 's main language. Under <code>polyglossia</code> , <code>hyperref</code> 's names stay English.
captions	<code>babel</code> 's main language, or <code>polyglossia</code> 's (<code>\setmainlanguage</code>), which do not always agree: “Table 1 –” under <code>babel</code> , “Tab. 1 :” under <code>polyglossia</code> , in French.

The names are measured, not translated: `tools/measure_names.py` compiles a probe document in each language and loading style, reads the names off the PDF, and checks the table the converter carries. English, French, German, Spanish, Italian, Portuguese and Dutch are measured. Another language prints English, and the run says so.

6.10 Differences from $\text{\LaTeX}$

- A reference to a list item whose label has parentheses keeps them: “2(b)” where $\text{\LaTeX}$ prints “2b”, “(ii)” for the `enumerate` package's `[(i)]` where $\text{\LaTeX}$ prints “ii”. Neither word processor has a reference format that drops them.
- References to equations are text, as decided for this project; they do not follow a renumbering.
- `\cpageref` with several labels on one page prints “pages 1 and 1” where `cleveref` merges them into “page 1”: the pages are known only after the phrase is written.
- A `table` environment with no `tabular` in it, an image in a table float, is dropped by `pandoc` along with its caption and label. A reference to it prints “??”.
- `enumitem`'s `label=` is not read; such a list is numbered as the article default.

7 The rest of the document

7.1 Citations

`natbib` and `biblatex` citations are resolved with `pandoc`'s `citeproc`, and a reference list is added at the end. The document already declares its bibliography with `\addbibresource` or `\bibliography`, and that is what is used:

```
linguexx2odt paper.tex          # the .bib the document names
linguexx2odt paper.tex --bibliography refs.bib # or another
linguexx2odt paper.tex --csl unified-style-sheet.csl
```

`\citet` comes out author-in-text, `\citep` parenthetical, and an optional locator survives: `\citep[786]{fruyt2011}` gives “Fruyt 2011, 786”. `Pandoc` has a single author-in-text mode, so `\citealt`, `\citeauthor` and `\citeyear` all print “Author (Year)”; the run says how many there are of each.

If no `.bib` is found, the keys are printed, “[vincent1982]”, and the run says so. A citation nothing resolved would otherwise disappear from the sentence.

7.2 Included files and macros

`\input`, `\include` and `\includeonly` are read and converted as part of the document, examples and references included. A file is looked for in the main document's directory, then through `kpsewhich`, so `TEXINPUTS` and `~/texmf` are searched too. A file the `TEX` distribution ships, such as `\input{glyphtounicode}`, is engine configuration: it is found and not read. Warnings name the file and line they are about.

The document's own macros (`\newcommand`, `\renewcommand`, `\providecommand`, `\def`) are expanded inside examples, arguments and optional defaults included. `\NewDocumentCommand` and `\def` with delimited parameters are not read.

7.3 What pandoc deletes

Anything the converter does not handle goes to pandoc. What pandoc cannot convert either is deleted from the output, and the run names each command or environment it deleted, with a count. Commands that set no text — spacing, page breaks, font sizes — are deleted without a word.

8 Writing examples in the word processor

A converted document often goes on growing in the word processor. Three companions build the converter's examples there: the same table, the same named styles, the same `NumEx` numbers. A converted example and a hand-built one are therefore the same kind of object, and both renumber together.

8.1 The Writer macro

Type an example as lines, select them, and run `Typeset example`:

```
Esto es un ejemplo glosado
this is a example glossed
'This is a glossed example.'
```

The result is a numbered, aligned example. The macro does two things the converter cannot: it *measures* the real font instead of estimating, and it reads the actual page instead of trusting `--text-width`.

- Sub-examples typed `a.`, `b.`; a leading `*` or `?` for a judgment; `{braces}` for one column; `\exannot{...}` for a label.
- `Typeset numbered tree` and `Typeset unnumbered tree` draw syntax trees from bracket notation, `[DP [D the] [NP [N tree]]]`, with roofs and movement arrows.
- `Untypeset example` gives an example back as the lines it was built from, with its number at their head. Edit them and typeset again, and it is the same example with the same number, so no reference breaks.
- `Example layout...` sets the indents and the spacing, which belong to the document.

Install it as a LibreOffice extension: add `linguexx-0.1.0.oxt` (from the project's latest release, or `python3 tools/build_oxt.py`) in Tools → Extension Manager. `docs/macro.md` describes it in full.

8.2 The Word and OnlyOffice add-ins

Two add-ins do in Word and OnlyOffice what the macro does in Writer: `Typeset selection`, `Typeset tree`, `Untypeset example`, `Example layout`, and `Insert reference...`, which lists the

document’s examples and inserts a live reference, “(3)” or the bare “3”.

- **OnlyOffice:** install `linguexx-onlyoffice.plugin` from the latest release (or `make onlyoffice`) through `Plugins → Plugin Manager`. Renumbering is immediate.
- **Word:** the add-in has to be served over HTTPS; for now that is a server on your own machine, `python3 tools/serve_addin.py`. Then upload `addin/word/manifest.xml` under `Home → Add-ins → My Add-ins`. Word for the desktop rennumbers with `Ctrl+A, F9`. Word on the web never rennumbers fields; the pane says which numbers are stale.

9 Back to L^AT_EX

A document that went to Word or Writer and was edited there can come back:

```
docx2linguexx paper.docx -o paper.tex
odt2linguexx paper.odt -o paper.tex
```

Every example table becomes a `linguexx` example again: tiers, braced groups, judgments, sub-examples on both levels, the translation and `\exannot`. Every reference to an example becomes `\ref` or `\pref` with a label. The rest is pandoc’s L^AT_EX writer, with `\usepackage[lazy]{linguexx}` added.

It reads the paragraph styles and fields that the converter, the macro and the add-ins write. An example typed by hand with tabs or spaces is not recognised and comes back as prose or as a table. An `.odt` is first saved as `.docx` by LibreOffice, which therefore has to be installed, because pandoc’s own ODT reader drops the text of every cross-reference.

The way back’s test is the way forward: every test document, and every document in `linguexx`’s own test suite, is converted to `.docx` and back, and must give the same examples, cell for cell.

- Labels are generated (`ex:1`, `ex:2a`): the word processor never had the author’s. Only an example something refers to gets one.
- Drawn trees come back as `forest` trees.
- The layout is not read back: `linguexx` sets its own widths and bands.

10 Limits

The converter warns about everything in this list.

Construct	What happens
<code>\ex.</code> inside <code>itemize</code> , <code>enumerate</code> or a footnote	not converted: the example’s text stays as ordinary unnumbered text
<code>\altn</code> , <code>\altg</code>	not supported: neither format can draw a stack of alternatives. Deleted with their arguments (printed as source inside an example in a <code>.docx</code>)
a list or table inside an example	not supported: an example table cannot hold a second layout
<code>\mvto</code> , <code>\mvfrom</code>	the arrows are not drawn; the words stay

Construct	What happens
forest, \Tree trees	kept as bracket notation, for the macro or the add-ins to draw
\exsource	set at the end of the example, not flush right
\GlossTransSide	the translation is set below, not beside: a side column has no defined behaviour when an example splits into bands
[phantomalign]	judgment marks get a column of their own
\GlossTierFont, \SetLeipzig, \DeclareJudgment	the styles and the literal marks are used
a reference to a sub-sub-example	printed without its sub-example's letter, “(1i)” for “(1b-i)”
[langsci]'s \ea... \z,	out of scope: the first degrades to unnumbered text,
gb4e's exe	the second is deleted
[legacy] mode	out of scope
math inside an example	handed to pandoc, which may not keep it

The cross-references' own differences are in section [6.10](#).

11 Troubleshooting

The numbers did not change after I inserted an example.

Update the fields: F9 in Writer, Ctrl+A then F9 in Word for the desktop. Word on the web does not update fields.

OnlyOffice shows an old number.

OnlyOffice shows a field's stored value. The converter stores the right one; an edit made since is shown once the field is refreshed.

A page reference shows “?”.

The conversion had no LibreOffice or `pdfinfo` to lay the document out. The word processor fills it in when it updates the fields.

A column is too narrow or too wide.

The widths are estimated (section [5.5](#)). Check `--font` and `--font-pt`, drag the column in the word processor, or rebuild the example with the macro, which measures.

`python3 -m linguexx2odt` finds no module.

Run it with `PYTHONPATH=src` from a checkout, or install the package.

A test or conversion that renders seems to hang.

Suspect OpenCL: LibreOffice probes it at startup, and a broken entry in `/etc/OpenCL/vendors` hangs the probe. `SAL_DISABLE_OPENCL=1` works around it.

12 For developers

```
make check      # lint and the test suite: what CI runs
make test      # the suite alone
```

```

make js-test          # the add-ins' suite (Node 22 or later)
make onlyoffice-test  # the OnlyOffice plugin, in Document Builder
make manual           # this manual

```

The end-to-end tests build real documents, render them with LibreOffice and measure the PDF. They check that numbers evaluate (with every stored value corrupted first, so that echoing cannot pass), that references resolve, that glosses sit under their words, that judgment marks do not shift the text and that nothing runs into the margin. They need `pandoc`, `soffice` and `pdftotext`. A missing one is a *failure*, not a skip, so that a contributor without LibreOffice does not see green and ship; `LINGUEXX2ODT_ALLOW_MISSING=1` runs a partial suite on purpose.

Several tables in the converter are measurements, each with a tool that checks it:
tools/measure_advances.py the per-character widths of the Times-metric faces, against the font;
tools/measure_names.py the names of references and captions, per language, against L^AT_EX;
tools/validate_docx.py a .docx against the ECMA-376 schemas;
tools/sync_macro.py the constants the converter shares with the macro and the add-ins.
notes/findings.md records what was measured and why each design decision was taken; the **plan-*.md** files are the work orders, with what was tested and what remains. The repository's layout is described at the end of **README.md**.

Licence

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linguexx itself is under the L^AT_EX Project Public License 1.3c, a different licence for a different kind of work. The converter only reads linguexx documents; it includes none of the package's code.